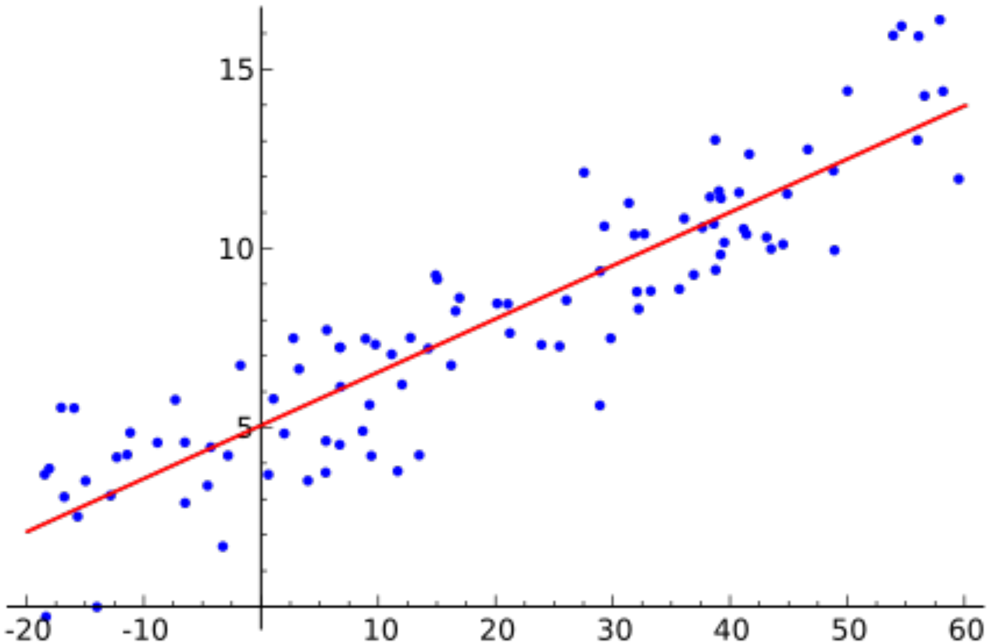
Homework 3
–
Linear Regression

Machine Learning – Simple Linear Regression

In this homework you will be familiarizing yourself with the basics of ***Simple Linear Regression***



source: wikipedia.org

What is simple linear regression? (1)

In statistics, linear regression is an approach for modeling the relationship between a scalar dependent variable y and one or more explanatory variables (or independent variables) denoted X. The case of one explanatory variable is called **simple linear regression**. For more than one explanatory variable, the process is called **multiple linear regression**. (This term should be distinguished from multivariate linear regression, where multiple correlated dependent variables are predicted, rather than a single scalar variable.)

In linear regression, the relationships are modeled using linear predictor functions (i.e. a line for a simple linear regression and a (hyper-)plane for multiple linear regression) whose unknown model parameters are estimated from the data. Such models are called linear models. Most commonly, the conditional mean of y given the value of X is assumed to be an affine function of X; less commonly, the median or some other quantile of the conditional distribution of y given X is expressed as a linear function of X. Like all forms of regression analysis, linear regression focuses on the conditional probability distribution of y given X, rather than on the joint probability distribution of y and X, which is the domain of multivariate analysis.

Linear regression was the first type of regression analysis to be studied rigorously, and to be used extensively in practical applications. This is because models which depend linearly on their unknown parameters are easier to fit than models which are non-linearly related to their parameters and because the statistical properties of the resulting estimators are easier to determine.

What is simple linear regression? (2)

Linear regression has many practical uses. Most applications fall into one of the following two broad categories:

- If the goal is prediction, or forecasting, or error reduction, linear regression can be used to fit a predictive model to an observed data set of y and X values. After developing such a model, if an additional value of X is then given without its accompanying value of y, the fitted model can be used to make a prediction of the value of y.
- Given a variable y and a number of variables $X_1, ..., X_p$ that may be related to y, linear regression analysis can be applied to quantify the strength of the relationship between y and the X_j, to assess which X_j may have no relationship with y at all, and to identify which subsets of the X_j contain redundant information about y.

Linear regression models are often fitted using the least squares approach, but they may also be fitted in other ways, such as by minimizing the "lack of fit" in some other norm (as with least absolute deviations regression), or by minimizing a penalized version of the least squares loss function as in ridge regression (L2-norm penalty) and lasso (L1-norm penalty). Conversely, the least squares approach can be used to fit models that are not linear models. Thus, although the terms "least squares" and "linear model" are closely linked, they are not synonymous. (Source: wikipedia.com)

Video 1: Simple Linear Regression (Part 1), The Very Basics

Estimated time: 30 min to watch + 20 min for the questions

- **Watch the video "Statistics 101: Simple Linear Regression (Part 1), The Very Basics“**
  https://www.youtube.com/watch?v=ZkjP5RJLQF4

- **Answer the following questions, perform any necessary calculations by hand on paper!**
  You are thinking about signing up for the class "Database Systems II", but before you do so, you would like to estimate the grade you can achieve in this class. You have interviewed 5 students who took the class last year and found out that the first student got a 1.0, the second a 3.0, der third a 2.0, the fourth a 1.3 and the fifth a 2.7.

  1) Given only this data, how would you estimate your expected grade? What grade do you expect?
  2) What is the difference between the residuals and the errors?
  3) Calculate the residuals! What is the sum of the residuals?
  4) What is the SSE and why do we use it? Calculate the SSE!

To verify your answers: 4) SSE=2,98

Video 2: Simple Linear Regression (Part 2), Algebra, Equations, and Patterns

Estimated time: 30 min to watch + 20 min for the questions

- **Watch the video** **"Statistics 101: Simple Linear Regression (Part 2), Algebra, Equations, and Patterns"**
  https://www.youtube.com/watch?v=iAgYLRy7e20

- **Answer the following questions**

  1) In the equation $y = m \cdot x + b$ for a line, what are m and b?
  2) In the video, you saw two very similar equations: $E(y) = \beta_0 + \beta_1 \cdot x$ and $\hat{y} = b_0 + b_1 \cdot x$. Explain the difference!
  3) In the regression equation $\hat{y} = b_0 + b_1 \cdot x$, what does it mean if the value for b_1 is zero? What if it is positive? What if it is negative?
  4) When conducting a simple linear regression with two variables, how do we determine how well the regression line fits the data?

Video 3: Simple Linear Regression (Part 3), The Least Squares Method

Estimated time: 30 min to watch + 50 min for the questions

- **Watch the video "Statistics 101: Simple Linear Regression (Part 3), The Least Squares Method"**
  You can skip the first 3 min; *https://www.youtube.com/watch?v=Qa2APhWjQPc*

- **Answer the following questions:**
  You are still thinking about signing up for the class "Database Systems II". You have gone back to the 5 students you interviewed who took the class last year (remember their grades: first student 1.0, second 3.0, third 2.0, fourth 1.3, fifth 2.7), and have asked them about their highschool grades ("Abiturnoten"): the first student had a 1.0, the second a 2.7, the third a 1.7, and fourth a 2.0 and the fifth a 3.3.

  1) What are the dependent and what are the independent variables?
  2) Draw a scatter plot for this data (by hand on paper)
  3) Compute the correlation! Is there a strong correlation? (If you do not remember correlation from your statistics classes, read up on it, e.g. at https://en.wikipedia.org/wiki/Correlation_and_dependence).
     Note: this questions involves slightly more calculations than all the other questions…
  4) Compute the centroid!
  5) Calculate b_1 and b_0 for the grade problem
  6) Add the regression to your scatter plot.
  7) Interpret the values for b_0 and b_1 you got!

To verify your answers: 3) 0,848918; 5) b0=0,24 and b1=0,822:

Congratulations!

You are now familiar with the basics of simple linear regression.

In the lecture, we will be looking at more scores and multiple linear regression (i.e., targets that depend on multiple features).